SECURE LAND SELLING AND OWNERSHIP VERIFICATION MANAGEMENT SYSTEM

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A Research Project Submitted to the School of Business and Economics in Partial Fulfilment of the Requirements for the award of Diploma in Business Information Technology, Kabarak University.

July, 2026

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DECLARATION AND APPROVAL

I declare that this work without any reasonable doubt has never been presented before to the School of Business and Economics or any other institution. No part of this research document shall therefore be duplicated without prior consent.

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Student: Naomi Jepkogei

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RECOMMENDATION

This research project entitled Secure Land Selling and Ownership Verification Management System written by Naomi Jepkogei is presented to the School of Business and Economics of Kabarak University. I have reviewed this project and recommended it to be accepted in partial fulfilment of the requirements for the Diploma in Business Information Technology.

Signature ……………………………………… Date …………………………….

…………………………………………,

School of Business and Economics,

Kabarak University.

ACKNOWLEDGEMENT

I thank God for the gift of life, good health and provision throughout my studies, without which I would not have reached this stage of my academic journey.

I am grateful to my supervisor and the lecturers of the School of Business and Economics for their guidance, patience and honest feedback during this project. Their input shaped both the system and this report. I also thank my family and classmates for their support and encouragement during the long hours this work demanded.

DEDICATION

This work is dedicated to my parents for their continuous support, guidance and sacrifice through my life and education, and to my fellow classmates for the assistance they offered throughout this study.

ABSTRACT

Land fraud remains one of the most damaging forms of economic crime in Kenya. Buyers lose savings to sellers who present forged title deeds, sell one parcel to several people, or sell land they do not own, because there is no simple way to confirm the facts of a parcel before money changes hands. This project designed and implemented LandGuard, a secure land selling and ownership verification management system. The system verifies the documents behind every listing, routes each parcel through a three-stage approval pipeline involving an administrator, a government land officer and a licensed surveyor, and scores every parcel for fraud risk using a machine-learning classifier before it can be listed. Payment is only possible after a government officer approves the transfer, and a completed sale automatically moves ownership, records the change in an ownership history, and issues a digitally signed certificate that anyone can verify by scanning a QR code without an account. The platform was built as three cooperating services: a web application developed with React and Next.js, a business interface developed with Node.js and Express with data held in a relational database through Prisma, and a fraud scoring service developed in Python. The system was developed using an Agile methodology and tested at unit, integration and system levels. Testing showed the fraud model reaching approximately ninety-nine percent accuracy on held-out data, and a complete sale was executed end to end, from offer through government approval and payment to certificate issue. The project demonstrates that a verification-first marketplace can make land transactions materially safer for ordinary buyers.

Keywords: land fraud, ownership verification, multi-stage approval, fraud scoring, digital certificate

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LIST OF ABBREVIATIONS

Abbreviation	Meaning
API	Application Programming Interface
DBIT	Diploma in Business Information Technology
DFD	Data Flow Diagram
ERD	Entity Relationship Diagram
HTTPS	Hypertext Transfer Protocol Secure
JWT	JSON Web Token
KES	Kenya Shillings
ML	Machine Learning
ORM	Object Relational Mapping
PIN	Personal Identification Number
QR	Quick Response (code)
RBAC	Role-Based Access Control
REST	Representational State Transfer
SQL	Structured Query Language
TOTP	Time-based One-Time Password
UAT	User Acceptance Testing
2FA	Two-Factor Authentication

CHAPTER ONE: INTRODUCTION

1.0 Background

Land is the most valuable asset most Kenyan families will ever own, and the basis of housing, farming and credit. Yet the process of buying land is one of the least protected transactions in the country. A buyer typically relies on photocopied title deeds, word of mouth and brokers whose interests are not always aligned with theirs. Official records exist, but checking them takes time, travel and money, and many buyers skip the step entirely. Fraudsters exploit this gap: the same parcel is sold to several buyers, forged title deeds are presented as genuine, and land is sold by people who do not own it (Boone, 2014).

The Government of Kenya has digitised parts of the land registry through the Ardhisasa platform, which allows online land searches for some counties (Ministry of Lands and Physical Planning, 2021). However, Ardhisasa records what the registry knows; it does not manage the sale itself. The moment of greatest risk, when money moves from buyer to seller, still happens outside any protective system. Research on land governance shows that secure, verifiable transactions raise investment and reduce disputes (Deininger & Feder, 2009). This project builds on that idea by putting verification, official approval and payment inside one controlled process.

1.1 Problem Statement

Land buyers in Kenya cannot easily confirm, before paying, that a parcel is genuine, that its documents are authentic, and that the seller is its true owner. Existing property websites advertise land but do not verify it; the registry can confirm a title but is not connected to the sale; and payment happens privately with no link to ownership transfer. The result is repeated fraud: double sales, forged deeds and sales by non-owners, with buyers discovering the problem only after their money is gone. There is no platform that verifies a parcel, controls the approval of its sale, holds payment until that approval is given, and then transfers ownership with proof a buyer can check independently.

1.2 Purpose of the Study

The purpose of this study was to design, develop and test a secure land selling and ownership verification management system that reduces land fraud by verifying every listing, enforcing government approval before payment, scoring fraud risk automatically, and issuing digitally verifiable proof of ownership after a completed sale.

1.3 General Objective

To develop a secure land selling and ownership verification management system for Kenya that reduces land fraud through verification, transparency, controlled transactions and verifiable ownership records.

1.3.1 Specific Objectives

i. To design a multi-stage verification workflow in which every land parcel is reviewed by an administrator, verified by a government land officer and approved by a surveyor before it can be listed for sale.

ii. To develop a fraud detection component that screens every parcel for duplicate title deeds, duplicate parcel numbers, forged or reused documents, blacklisted sellers and abnormal pricing, and assigns a risk score using a machine-learning classifier.

iii. To implement a controlled transaction process in which payment is only possible after government approval, and a successful payment automatically transfers ownership and issues a digitally signed certificate verifiable through a QR code.

iv. To evaluate the system through unit, integration, system and user acceptance testing.

1.4 Research Questions

i. How can a multi-stage verification workflow prevent unverified land parcels from reaching buyers?

ii. Which fraud signals can be detected automatically from a parcel, its documents and its seller, and how accurately can a classifier score them?

iii. How can payment, ownership transfer and proof of ownership be bound together so that a buyer cannot lose money to an unapproved sale?

iv. To what extent does the developed system meet its functional and non-functional requirements under testing?

1.5 Justification of the Study

Land fraud causes direct financial loss to families, discourages investment and floods courts with ownership disputes. Transparency International (2019) reports that land services are among the public services where citizens most frequently encounter corruption in Africa. A system that verifies parcels before listing, blocks high-risk sellers automatically and gives buyers independent proof of ownership addresses the problem at the point where losses actually occur: the transaction. The project also demonstrates to the business-technology field how machine-learning scoring, workflow control and digital signatures can be combined in one practical product.

1.6 Scope of the Study

The system covers the full life of a land sale on one platform: seller registration, parcel listing, document upload, fraud screening, three-stage verification, marketplace search, offers, government approval of transfers, payment through mobile money, card or bank transfer, ownership transfer, certificate issue and public QR verification. It serves five user roles: buyer, seller, administrator, government land officer and surveyor. The system was developed and tested with simulated payment providers and a seeded demonstration registry; it does not replace the official national land registry and does not process live payments in its test configuration.

1.7 Limitations of the Study

i. The system relies on the accuracy of documents presented to it; a document that is forged so well that it matches registry records cannot be detected without direct registry integration.

ii. Payments run against provider sandboxes rather than live money movement, since live M-Pesa and card credentials require commercial agreements.

iii. The fraud classifier was trained on generated, domain-informed data because no public dataset of labelled Kenyan land-fraud cases exists.

iv. Testing was carried out with demonstration users rather than a live public deployment.

1.8 Assumptions of the Study

i. Government land officers and surveyors are willing to perform their approval steps through the platform.

ii. Sellers can produce scanned copies of their title documents.

iii. Users have access to a smartphone or computer with an internet connection.

iv. The information in uploaded documents reflects the official registry position at the time of listing.

CHAPTER TWO: LITERATURE REVIEW

2.0 Introduction

This chapter reviews the problem of land fraud, examines existing systems that touch the land transaction process in Kenya and elsewhere, identifies the gaps those systems leave open, and presents the conceptual framework that guided the design of LandGuard.

2.1 Land Fraud and Land Governance

Secure property rights are strongly linked to investment, credit access and economic growth, and weak land administration is a recognised brake on development (Deininger, 2003). Where registration systems are incomplete or slow, informal transactions dominate and fraud follows. Deininger and Feder (2009) show that land registration only delivers its benefits when records are trustworthy and cheap to check; a registry that is hard to consult pushes buyers back to informal assurances. In Kenya, land has a long history as an instrument of political patronage, which further complicates trust in ownership records (Boone, 2014). Surveys of citizens across Africa consistently place land services among the interactions where bribery is most common (Transparency International, 2019). The literature therefore points to a clear requirement: verification must be cheap, fast and available at the moment of the transaction.

2.2 Related Works

2.2.1 Ardhisasa (National Land Information Management System)

Ardhisasa is the Kenyan government platform for land services, offering online land searches, transfers and other registry functions for supported counties (Ministry of Lands and Physical Planning, 2021). It is authoritative where coverage exists, but it is a registry interface rather than a marketplace: it does not host listings, does not screen sellers, does not manage offers or payments, and cannot stop a buyer from paying for a parcel that was never checked.

2.2.2 e-Citizen Land Search

The e-Citizen portal provides official land search certificates for a fee. A diligent buyer can use it to confirm a title before purchase. The step is manual and optional, however, and nothing connects the search result to the payment that follows. A buyer can receive a clean search on Monday and still be defrauded on Tuesday by a seller who has sold the same parcel to someone else in the meantime.

2.2.3 Private Property Portals

Commercial portals such as BuyRentKenya and Property24 advertise land and houses at scale. They solve discovery, not trust: listings are posted by sellers and agents without document verification, the portals take no part in the transaction, and fraudulent listings are removed only after complaints. The buyer carries the entire verification burden.

2.2.4 Blockchain Land Registry Initiatives

Several countries have piloted distributed-ledger records for land, with the aim of making ownership entries tamper-evident. These pilots confirm strong demand for verifiable records, but they focus on the storage layer of the registry itself and typically require national-scale institutional change before citizens see any benefit. They do not address listing verification, seller screening or payment control in the sale process.

2.3 Gaps in Existing Systems

Table 2.1: Comparison of existing systems against transaction-security needs

Capability	Ardhisasa	e-Citizen search	Property portals	LandGuard
Hosts land listings	No	No	Yes	Yes
Verifies documents before listing	N/A	No	No	Yes
Screens sellers for fraud	No	No	No	Yes
Government approval inside the sale	No	No	No	Yes
Payment held until approval	No	No	No	Yes
Automatic ownership transfer record	Registry only	No	No	Yes
Public proof a buyer can verify	Search fee	Search fee	No	Free QR check

The pattern across all existing systems is separation: discovery, verification, approval and payment each happen in different places, and fraud lives in the gaps between them. No reviewed system verifies a listing before publication, blocks risky sellers automatically, gates payment behind official approval and issues independently checkable proof of ownership. LandGuard was designed to close exactly these gaps by joining the four steps in one controlled workflow.

2.4 Conceptual Framework

The study is guided by the framework below. The independent variables are the four controls the system introduces; acting together through the platform, they produce the dependent outcomes of reduced fraud and verified, transparent transfers.

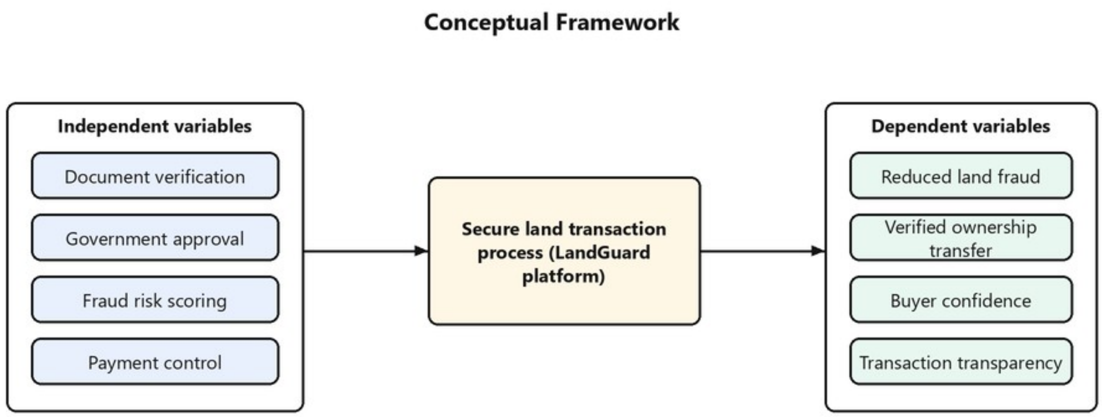

Figure 2.1: Conceptual framework of the study

CHAPTER THREE: RESEARCH METHODOLOGY

3.0 Introduction

This chapter describes the methodology used to develop the system, the functional and non-functional requirements it was built to meet, the tools used, the way data was collected and analysed, and the milestones that structured the work.

3.1 System Development Methodology

The system was developed using the Agile methodology, working in short iterations that each delivered a usable increment: first authentication and user roles, then parcel management and documents, then the verification pipeline, then the fraud engine, then transactions and payments, and finally certificates and QR verification. Each iteration ended with testing and a review of the running software, and feedback from each review shaped the next iteration (Beck et al., 2001). Agile suited this project because the requirements became clearer as working software exposed real workflow questions, such as when exactly a fraud scan should block a submission (Sommerville, 2016).

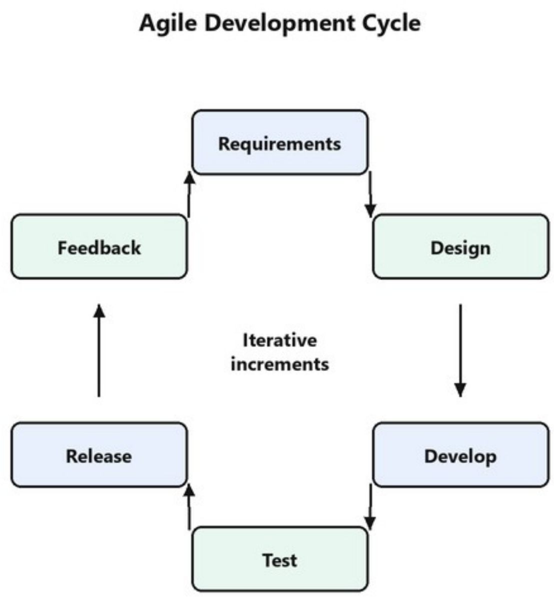

Figure 3.1: Agile development cycle used in the project

3.1.1 Justification of the Methodology

A waterfall approach would have required the full fraud-detection design to be fixed before any code existed, yet the most important design decisions, such as storing duplicate title deeds so they can be detected rather than rejecting them at the database level, only became visible once the first version of the fraud engine ran against real records. Agile allowed those discoveries to improve the design instead of breaking a fixed plan.

3.2 Functional Requirements

Table 3.1: Functional requirements of the system

Code	Requirement
FR1	The system shall register users as buyers or sellers, and allow administrators to provision officers, surveyors and other administrators.
FR2	The system shall authenticate users with email and password, support two-factor authentication, and enforce role-based permissions on every action.
FR3	The system shall allow sellers to create land parcels with location, size, land use, price and

Code	Requirement
	description, and to upload supporting documents.
FR4	The system shall fingerprint every uploaded document and detect documents reused across different sellers.
FR5	The system shall screen every submitted parcel for fraud signals and compute a risk score from zero to one hundred, blocking parcels that score sixty or more.
FR6	The system shall route each parcel through administrator review, government verification and survey approval before listing it.
FR7	The system shall allow buyers to search verified listings and make offers, and sellers to accept or reject offers.
FR8	The system shall require government officer approval of an accepted offer before any payment can be made.
FR9	The system shall accept payment through M-Pesa, Visa, MasterCard or bank transfer.
FR10	The system shall, on successful payment, transfer ownership, record the change in an ownership history, and issue a digitally signed certificate with a QR code.
FR11	The system shall allow anyone to verify a certificate or parcel by QR code without an account.
FR12	The system shall record every security-relevant action in an audit trail and deliver notifications to affected users.

3.3 Non-functional Requirements

Table 3.2: Non-functional requirements of the system

Category	Requirement
Security	Passwords stored as bcrypt hashes; JWT access tokens with rotating refresh tokens; optional TOTP two-factor login; input validation on every endpoint; rate limiting; signed certificates and QR payloads.
Usability	Role-specific dashboards; responsive layout usable on phones and desktops; public verification requires no account or training.
Reliability	Multi-step operations such as ownership transfer run inside database transactions so they either complete fully or not at all; fraud scoring falls back to a rule engine if the model service is unavailable.
Performance	Interface responses within a few hundred milliseconds under normal load; production page loads measured in milliseconds.
Portability	Identical schema runs on SQLite in development and PostgreSQL in production; the whole platform can be started with a single command or through containers.
Auditability	Every important action is written to an append-only audit log with actor, entity and timestamp.

3.4 Tools and Techniques

Table 3.3: Tools used in developing the system

Tool	Use in the project
React and Next.js	Building the web application and the five role dashboards.
Node.js and Express	Building the business interface that holds the rules and data access.
TypeScript	The language of both the web application and the business interface.
Prisma ORM	Defining the database schema and generating safe, typed queries.
SQLite and PostgreSQL	The development and production databases respectively.
Python and Flask	The fraud scoring service hosting the logistic-regression classifier.
Socket.IO	Real-time delivery of messages and notifications.
Visual Studio Code	Writing and debugging the source code.
Git	Version control of the source code.
Microsoft Word	Preparing this project documentation.

3.5 Milestones and Deliverables

Table 3.4: Project milestones and deliverables

Milestone	Deliverable
Requirements and design	Requirement lists, workflow design, database design and diagram set.
Core platform	Authentication, role-based access, user management and dashboards.
Land and verification	Parcel management, document upload with fingerprinting, three-stage verification pipeline.
Fraud engine	Rule checks, feature extraction, trained classifier and scoring service.
Transactions	Offers, government approval, payments, ownership transfer, certificates and QR verification.
Testing and documentation	Test results, user manual, deployment guide and this report.

3.6 Data Collection Methods

3.6.1 Primary Data Collection

Primary information about how land sales go wrong was gathered through informal interviews with land buyers and a review of the manual steps a careful buyer performs today: obtaining a search, visiting the parcel, consulting the area land control board and paying a deposit. These steps were mapped and each became either a feature of the system or a check inside its workflow.

3.6.2 Secondary Data Collection

Secondary data came from published research on land governance and registration (Deininger, 2003; Deininger & Feder, 2009; Boone, 2014), reports on corruption in land services (Transparency International, 2019), and the public documentation of the Ardhisasa platform and the payment and development tools used (Ministry of Lands and Physical Planning, 2021; Safaricom, 2024).

3.6.3 Data Analysis

Collected fraud patterns were analysed into nine measurable signals per parcel, such as duplicate title deed counts and deviation from local market price. These signals became the features of the fraud classifier, and a labelled training set of four thousand examples was generated from the documented patterns to train and evaluate it. Model quality was analysed with accuracy, precision, recall and F1 on a held-out twenty percent test split.

3.7 Data Flow Diagrams

3.7.1 Context Diagram (Level 0)

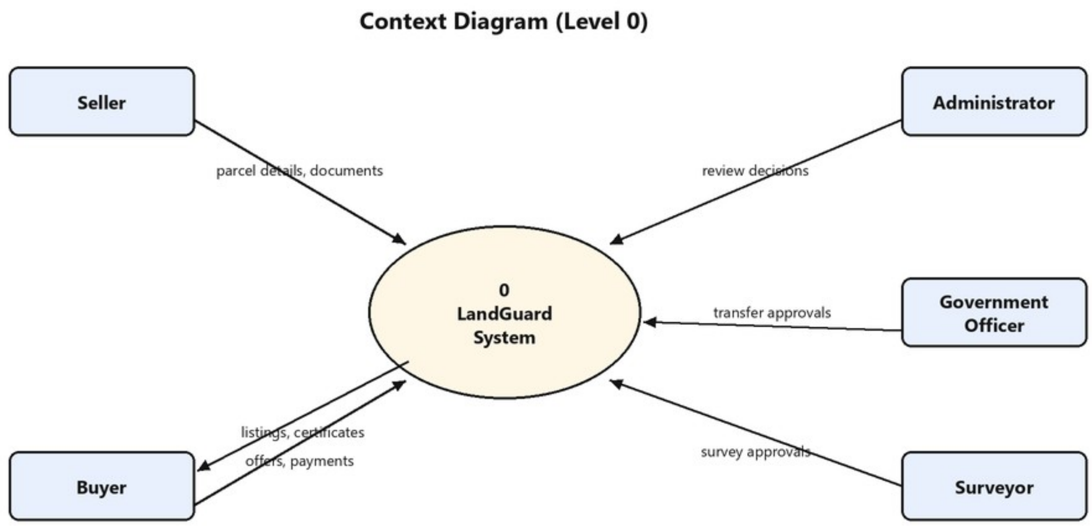

Figure 3.2: Context diagram of the system

At context level, five external entities interact with the system: sellers supply parcel details and documents; buyers search, make offers and pay; administrators, government officers and surveyors return review, approval and survey decisions; and buyers and the public receive listings and verifiable certificates.

3.7.2 Data Flow Diagram (Level 1)

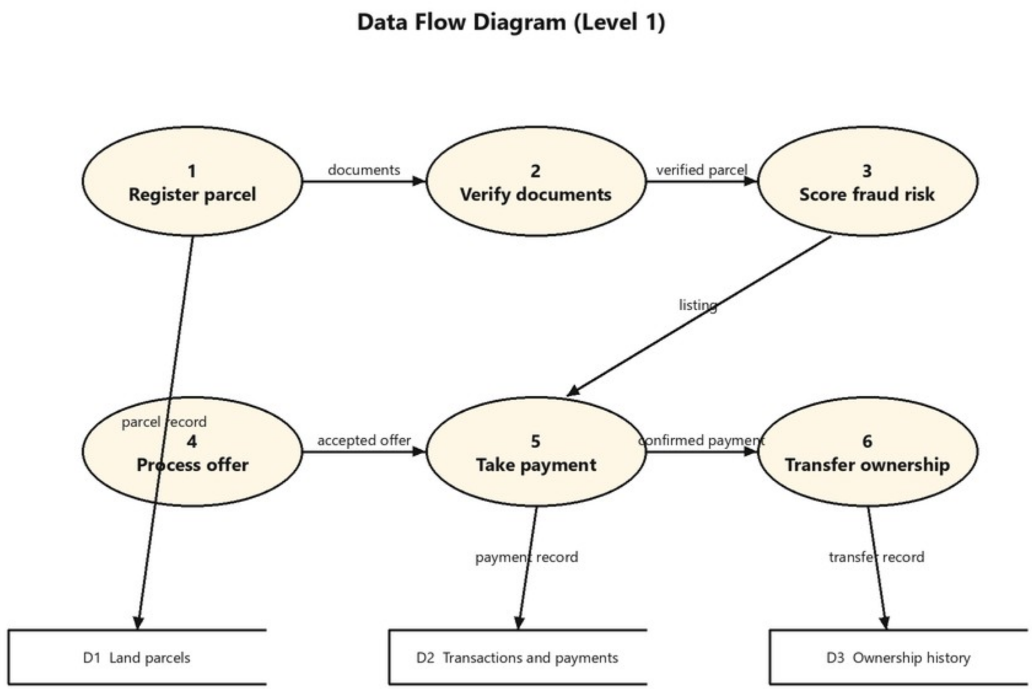

Figure 3.3: Level one data flow diagram

Level one expands the system into six processes. Registration feeds document verification, which feeds fraud scoring; verified parcels flow to offer processing, payment and finally ownership transfer. Three data stores hold parcels, transactions with payments, and the ownership history.

CHAPTER FOUR: SYSTEM ANALYSIS, IMPLEMENTATION AND DESIGN

4.0 Introduction

This chapter presents the analysis and design of the system and describes the implementation as it was actually built. All diagrams, schema tables and test results in this chapter reflect the running software. The complete source code of the system is hosted on GitHub at https://github.com/Ambroise2/landguard (to be published; see submission checklist).

4.1 System Architecture

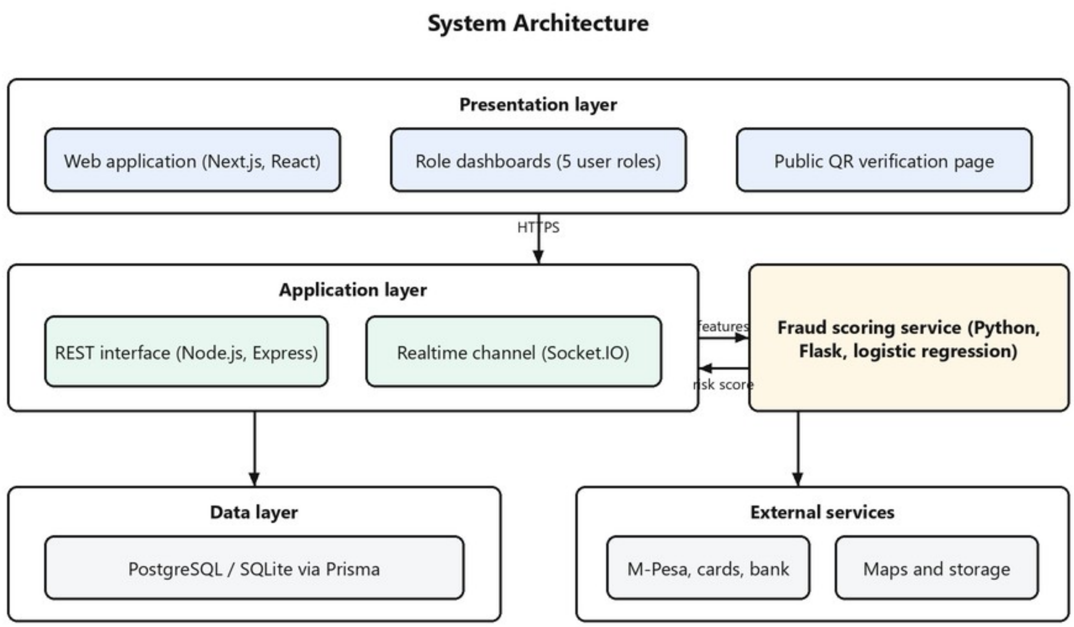

Figure 4.1: System architecture

The platform is organised as three cooperating services. The presentation layer is a Next.js web application serving the five role dashboards and the public QR verification page. The application layer is an Express interface that holds all business rules, authentication and data access, together with a Socket.IO channel that pushes notifications and messages in real time. A separate Python service hosts the fraud classifier; the interface sends it a feature vector for each parcel and receives a risk score with an explanation. Data is held in a relational database reached through the Prisma ORM, with SQLite in development and PostgreSQL in production. Payment providers, maps and file storage sit behind small adapter interfaces so a sandbox and a live provider can be swapped without changing business logic.

Separating the fraud model into its own service has a practical reason: the model can be retrained, scaled or replaced independently of the main platform, and if it is ever unreachable the interface falls back to a built-in rule-based score, so fraud screening never fails outright.

4.2 System Workflow

Every parcel follows one controlled path: the seller creates the parcel and uploads documents; submission triggers a fraud scan which can block the parcel immediately; an administrator reviews it; a government land officer verifies it; a surveyor approves it; only then is it listed. A buyer makes an offer, the seller accepts, a government officer approves the transfer, and only after that approval can the buyer pay. Successful payment transfers ownership, writes the ownership history and issues the certificate. Ownership therefore cannot change hands on an unverified parcel or an unapproved sale.

4.3 Analysis Diagrams

4.3.1 Use Case Diagram

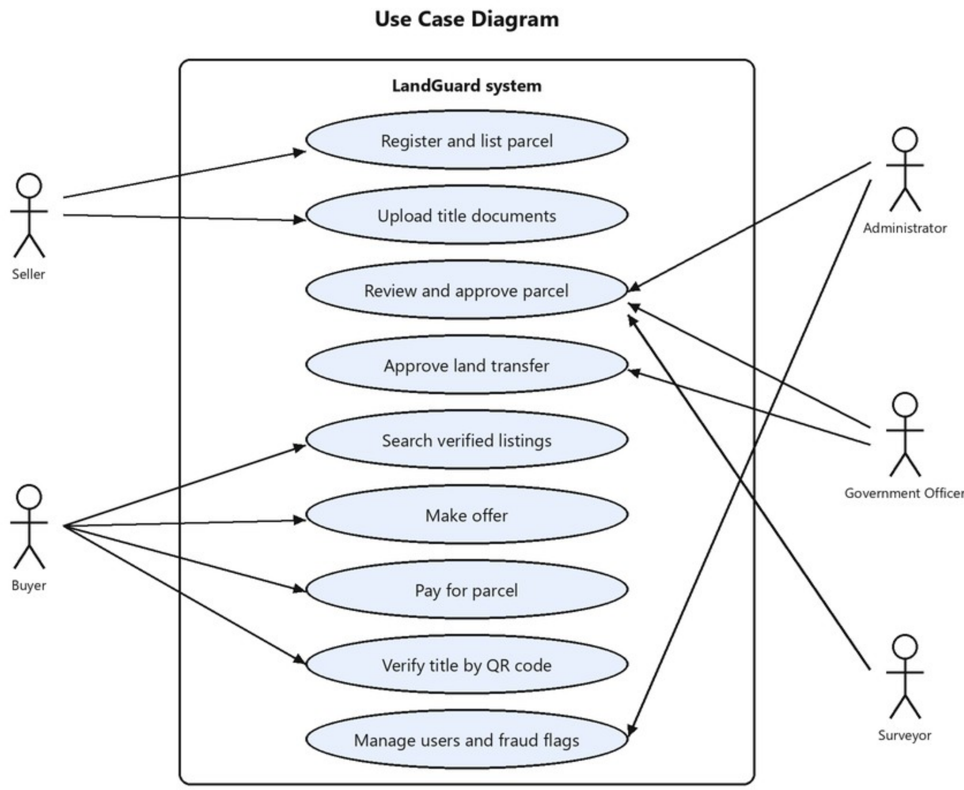

Figure 4.2: Use case diagram of the system

Sellers register parcels and upload documents. Buyers search, offer, pay and verify titles. Administrators review parcels and manage users and fraud flags. Government officers verify parcels and approve transfers. Surveyors give the final pre-listing approval. QR verification is also available to the general public without an account.

4.3.2 Sequence Diagram

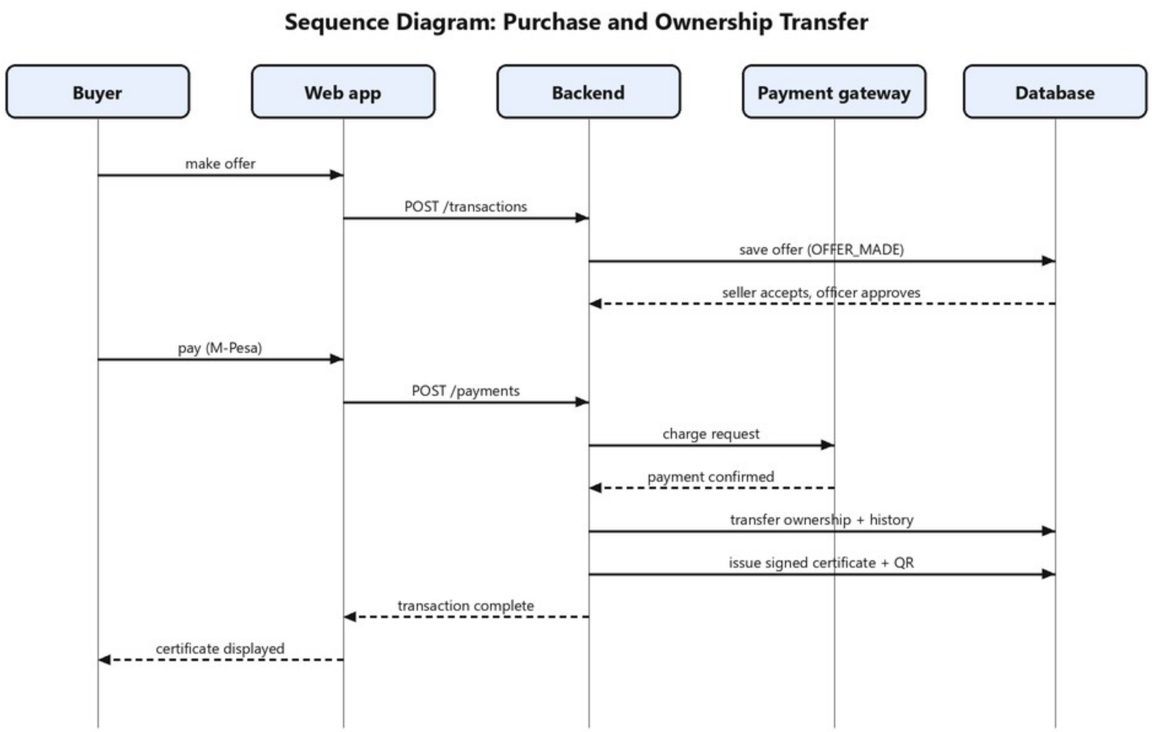

Figure 4.3: Sequence diagram for purchase and ownership transfer

The sequence shows the purchase flow. The offer is stored and moves through seller acceptance and government approval. The payment request goes to the selected provider; on confirmation the interface performs the settlement inside one database transaction: ownership moves to the buyer, the history entry is written, and the signed certificate with its QR code is created. The buyer sees the certificate on the transaction page.

4.3.3 Class Diagram

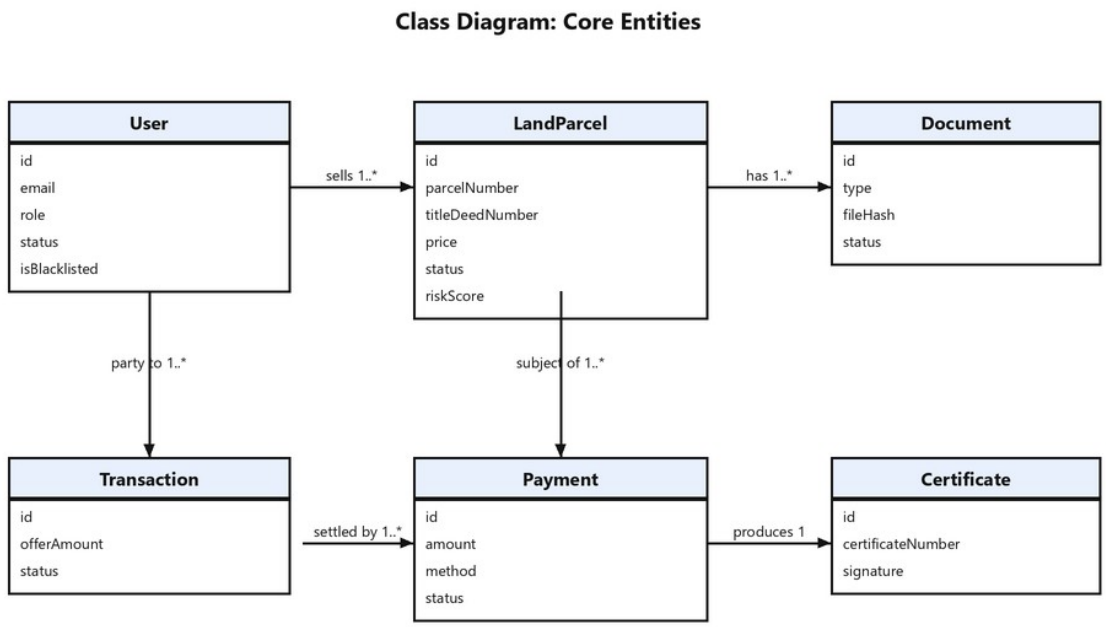

Figure 4.4: Class diagram of the core entities

4.3.4 Activity Diagram

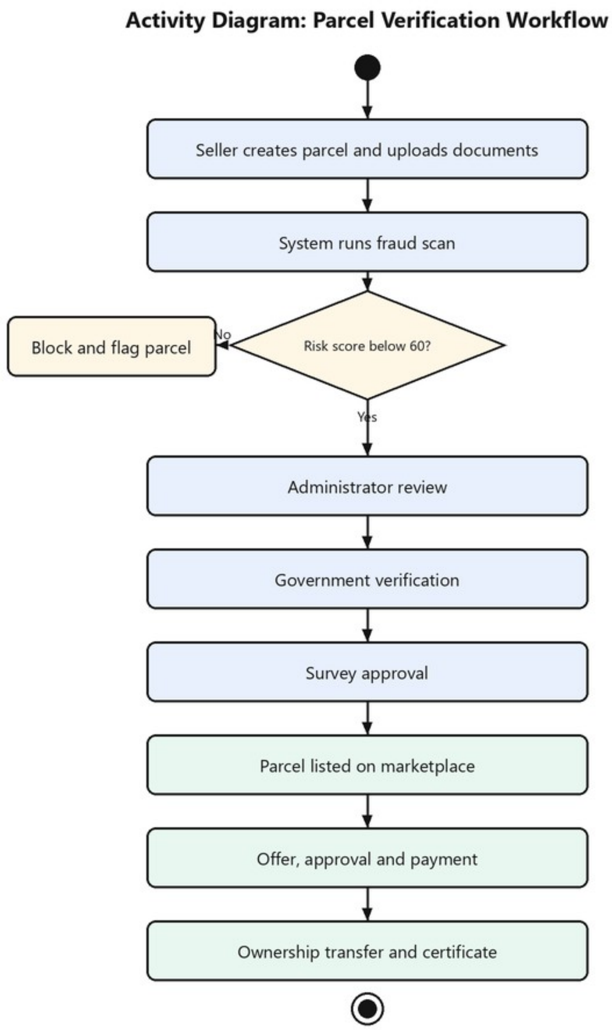

Figure 4.5: Activity diagram of the verification workflow

4.3.5 State Chart Diagram

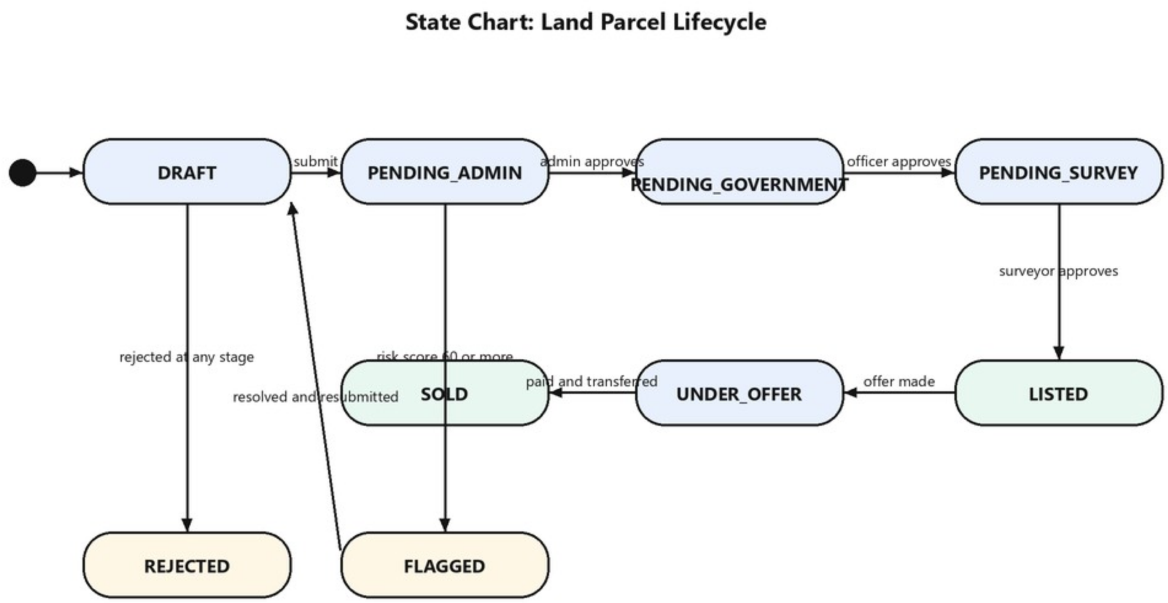

Figure 4.6: State chart of the land parcel lifecycle

A parcel starts as a draft and moves through the three pending states as each approval is given. Rejection at any stage returns it to the seller with notes; a fraud score of sixty or more sends it to the flagged state, from which it can only leave after investigation. Listing, offer, payment and transfer complete the lifecycle at the sold state.

4.3.6 Flowchart of Fraud Risk Scoring

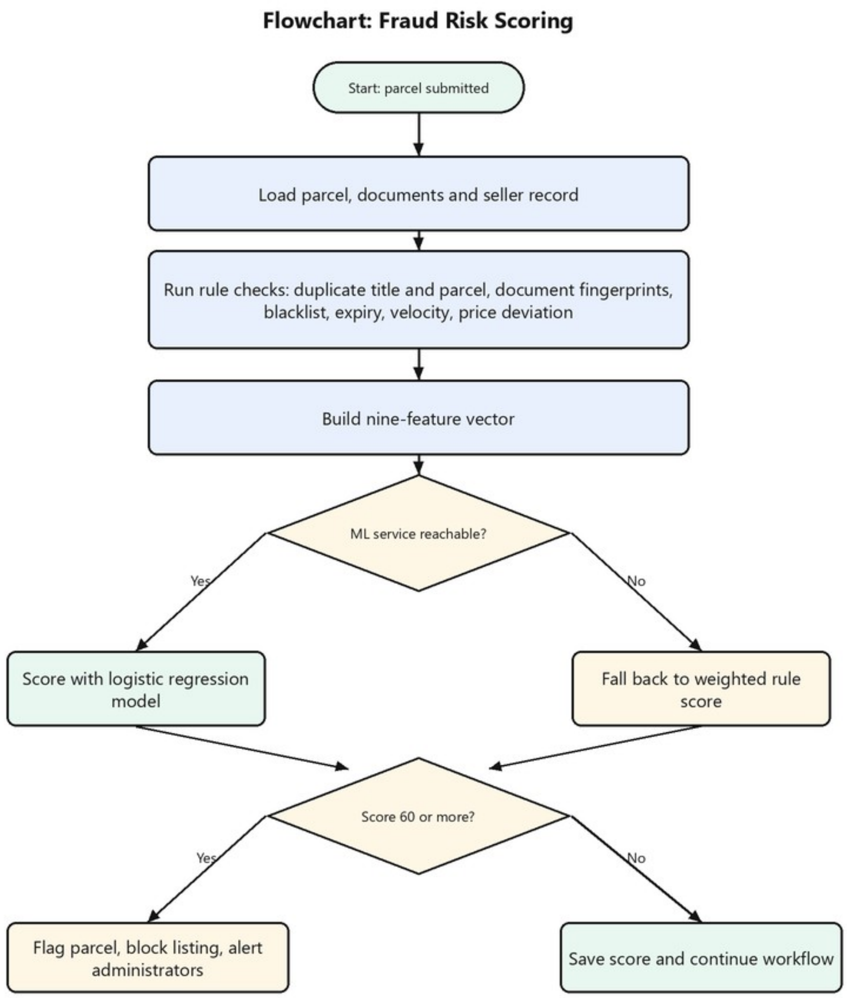

Figure 4.7: Flowchart of the fraud scoring process

4.4 Database Design

4.4.1 Entity Relationship Diagram

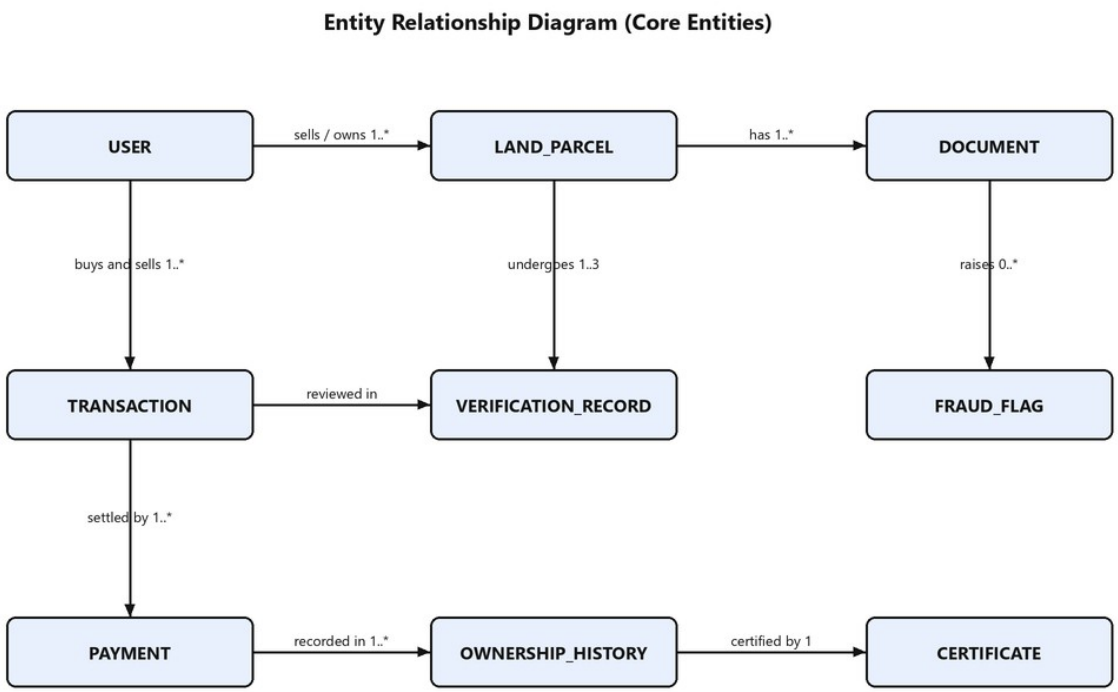

Figure 4.8: Entity relationship diagram of the core entities

4.4.2 Database Schema

The database holds seventeen tables. Status and role values are stored as validated text so the identical schema runs on both SQLite and PostgreSQL. Table 4.1 summarises all seventeen entities; the tables that follow detail the fields of the six central ones.

Table 4.1: All database entities and their purpose

Entity	Purpose
Role	Definition of the five roles and their permission sets.
User	Every account: buyers, sellers, administrators, officers and surveyors.

Entity	Purpose
RefreshToken	Long-lived login tokens, stored so they can be rotated and revoked.
LandParcel	A parcel offered on the platform, its location, price, workflow status and risk score.
Document	Metadata and fingerprint of each uploaded document.
VerificationRecord	One record per verification stage per parcel, with the decision and reviewer.
Transaction	A sale in progress from offer to completion.
Payment	Each payment attempt, its method, provider reference and outcome.
OwnershipHistory	The permanent chain of title for every parcel.
Certificate	The signed digital certificate issued after a completed sale.
Message	Direct messages between users about a parcel.
Notification	In-app alerts pushed to users in real time.
Complaint	Reports filed by users and their investigation status.
SiteVisit	Requests for physical inspection and the surveyor assigned.
FraudFlag	Each fraud signal raised, its severity, weight and description.
QrCode	Signed QR verification tokens for parcels and certificates.
AuditLog	The append-only record of every security-relevant action.

Table 4.2: User table (key fields)

Field	Type	Description
id	String (PK)	Unique identifier.
email / phone	String (unique)	Login email and contact number.
passwordHash	String	Password stored as a bcrypt hash, never in plain text.
role	String	BUYER, SELLER, ADMIN, GOVERNMENT_OFFICER or SURVEYOR.
status	String	ACTIVE, SUSPENDED or PENDING.
twoFactorEnabled / twoFactorSecret	Boolean / String	Two-factor login state and its TOTP secret.
isBlacklisted / blacklistReason	Boolean / String	Fraud blacklist flag and the recorded reason.

Table 4.3: LandParcel table (key fields)

Field	Type	Description
id	String (PK)	Unique identifier.
parcelNumber /	String	Registry identifiers. Deliberately not database-

Field	Type	Description
titleDeedNumber		unique: duplicates must be storable so the fraud engine can detect and flag them.
county / subCounty / locality	String	Location of the land.
sizeAcres / landUse / price	Float / String / Float	Size, permitted use and asking price in KES.
status	String	DRAFT, PENDING_ADMIN, PENDING_GOVERNMENT, PENDING_SURVEY, VERIFIED, LISTED, UNDER_OFFER, SOLD, REJECTED or FLAGGED.
riskScore	Int	Fraud risk from 0 to 100, written by the scoring engine.
sellerId / currentOwnerId	String (FK)	The listing seller and the recorded current owner.

Table 4.4: Document table (key fields)

Field	Type	Description
id	String (PK)	Unique identifier.
parcelId	String (FK)	The parcel the document supports.
type	String	TITLE_DEED, SURVEY_MAP, NATIONAL_ID, LAND_SEARCH, CONSENT or RATES_CLEARANCE.
fileHash	String	SHA-256 fingerprint of the file; identical fingerprints from different sellers expose reused or forged documents.
status	String	PENDING, VERIFIED, REJECTED or EXPIRED.
expiryDate	DateTime	Expiry for time-limited documents such as land searches.

Table 4.5: Transaction table (key fields)

Field	Type	Description
id / reference	String	Unique identifier and human-readable reference.
parcelId / buyerId / sellerId	String (FK)	The parcel and the two parties.
offerAmount	Float	The amount offered in KES.
status	String	OFFER_MADE, GOV_APPROVAL_PENDING, PAYMENT_PENDING, PAID, COMPLETED, REJECTED or CANCELLED.

Field	Type	Description
govApprovedById	String (FK)	The officer who approved the transfer.

Table 4.6: Payment and Certificate tables (key fields)

Field	Type	Description
Payment.method	String	MPESA, VISA, MASTERCARD or BANK_TRANSFER.
Payment.status	String	PENDING, SUCCESS, FAILED or REFUNDED.
Payment.providerRef	String	Reference returned by the payment provider.
Certificate.certificateNumber	String (unique)	The printed certificate number, for example LG-CERT-2026-CMRHX48D.
Certificate.signature	String	HMAC signature proving the certificate was issued by the platform and has not been altered.
Certificate.qrCodeId	String (FK)	The QR code that verifies this certificate.

4.5 Authentication and Access Control

Login issues a short-lived access token and a refresh token that rotates on every use; a stolen refresh token dies the first time it is replayed. Users can enable two-factor authentication with any standard authenticator application, after which login requires the six-digit code. Every endpoint checks the caller against a permission map derived from the five roles, so a buyer physically cannot call an approval endpoint, and only administrators can manage users or blacklists. All authentication events are written to the audit trail with the actor, address and outcome.

4.6 Fraud Detection and the Machine-Learning Model

Fraud detection runs in two layers. The rule layer inspects a submitted parcel and records human-readable flags: a title deed number that appears on another listing, a document byte-identical to one filed by a different seller, a blacklisted seller, expired documents, a missing title deed, abnormal listing velocity, a price far below the county average, or a seller who is not the recorded owner. The same inspection produces a nine-feature numeric vector that is sent to the scoring service.

The scoring service hosts a logistic-regression classifier implemented from first principles and trained by gradient descent with L2 regularisation on four thousand labelled examples (Hosmer, Lemeshow, & Sturdivant, 2013). Implementing the mathematics directly keeps every prediction explainable, which matters for a score that can block a land sale. On the held-out test split the model reached 98.9 percent accuracy, 99.6 percent precision, 97.1 percent recall and an F1 score of 0.98. Each response returns the probability, the zero-to-one-hundred score and the features that pushed the score up or down. A parcel scoring sixty or more is blocked from listing, flagged and reported to administrators automatically.

4.7 Transactions, Payments and Certificates

An accepted offer cannot be paid until a government officer approves the transfer, which is the system's central safeguard: money cannot move on an unapproved sale. Payments run through a single gateway interface with M-Pesa, Visa, MasterCard and bank transfer implementations. When a payment succeeds, settlement executes atomically: ownership moves to the buyer, the ownership history gains a new entry, the transaction completes, and a certificate is issued carrying an HMAC signature and a QR code. Scanning the code opens a public page that confirms the certificate is genuine and shows the parcel and its current owner; the check recomputes the signature server-side, so a tampered certificate fails immediately.

4.8 Implementation Screens

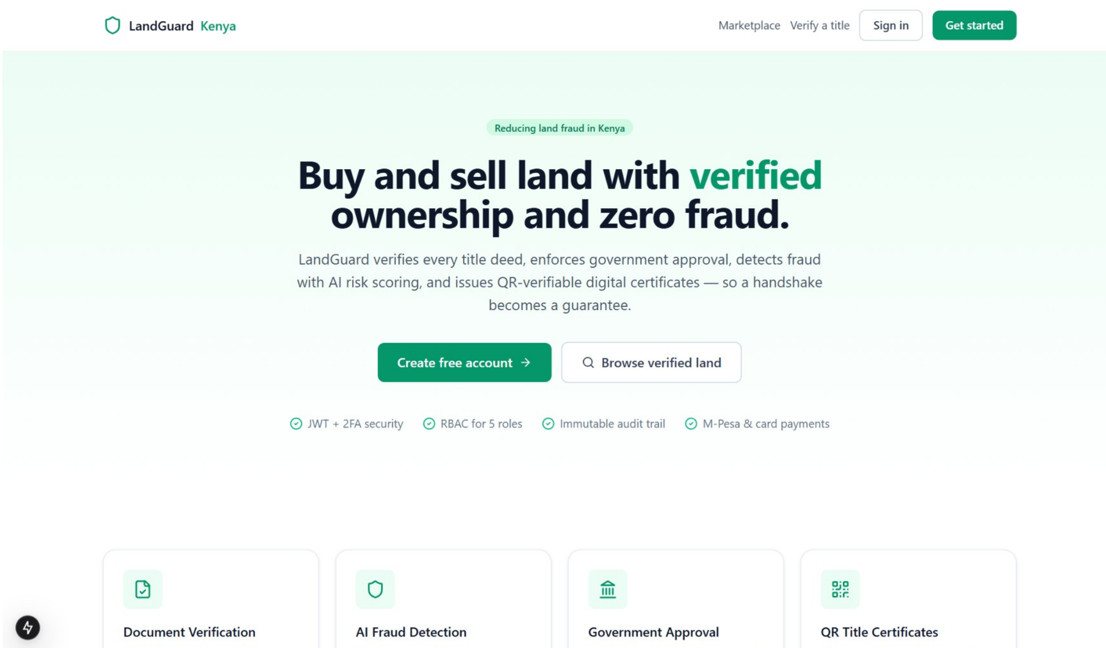

Figure 4.9: Landing page

The figure above shows the public landing page. It introduces the platform, states the verification guarantees, and offers entry points to the marketplace, registration and title verification.

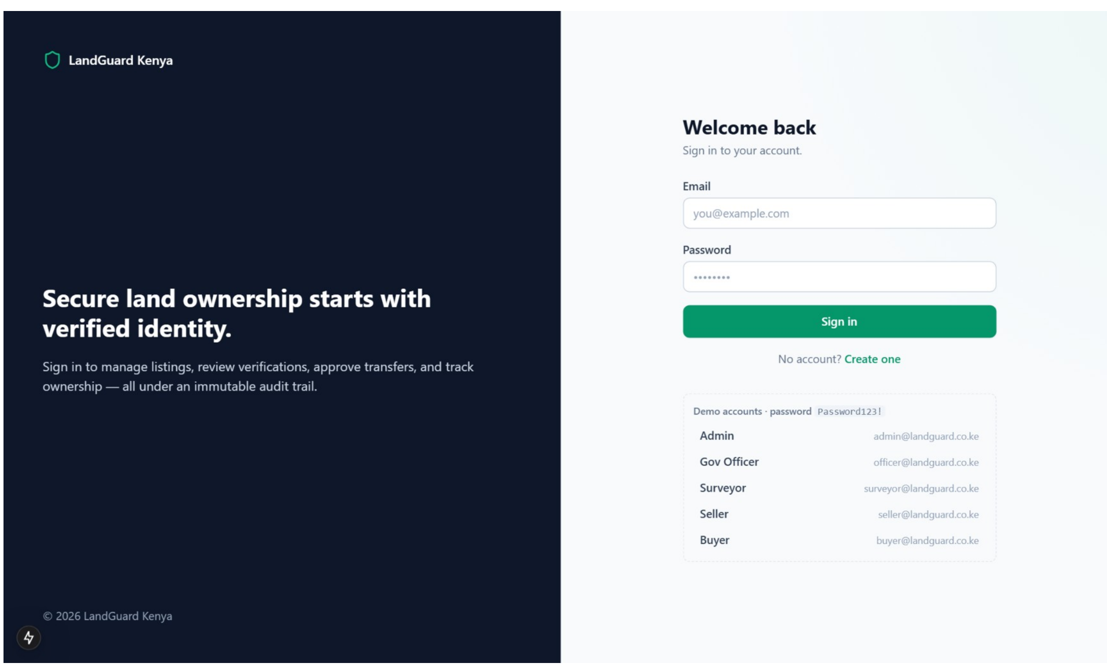

Figure 4.10: Login page

The figure above shows the login page. Users sign in with email and password; accounts with two-factor authentication enabled are prompted for their six-digit code before access is granted.

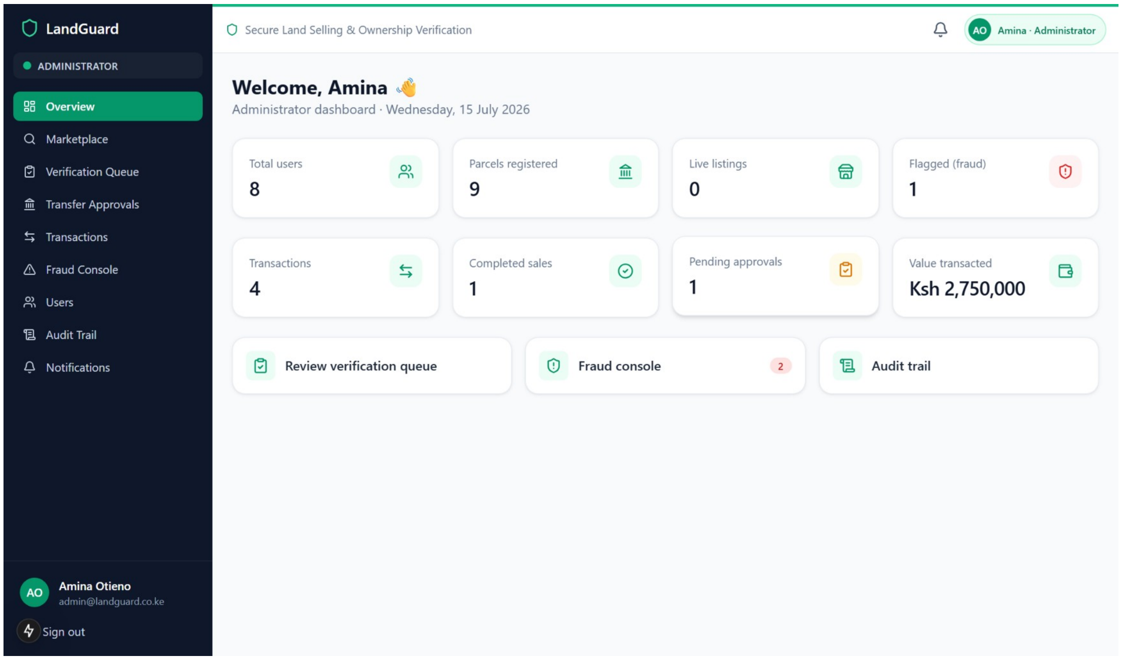

Figure 4.11: Administrator dashboard

The figure above shows the administrator dashboard with live platform indicators: registered users, parcels, listings, flagged parcels, transactions, pending approvals and total value transacted.

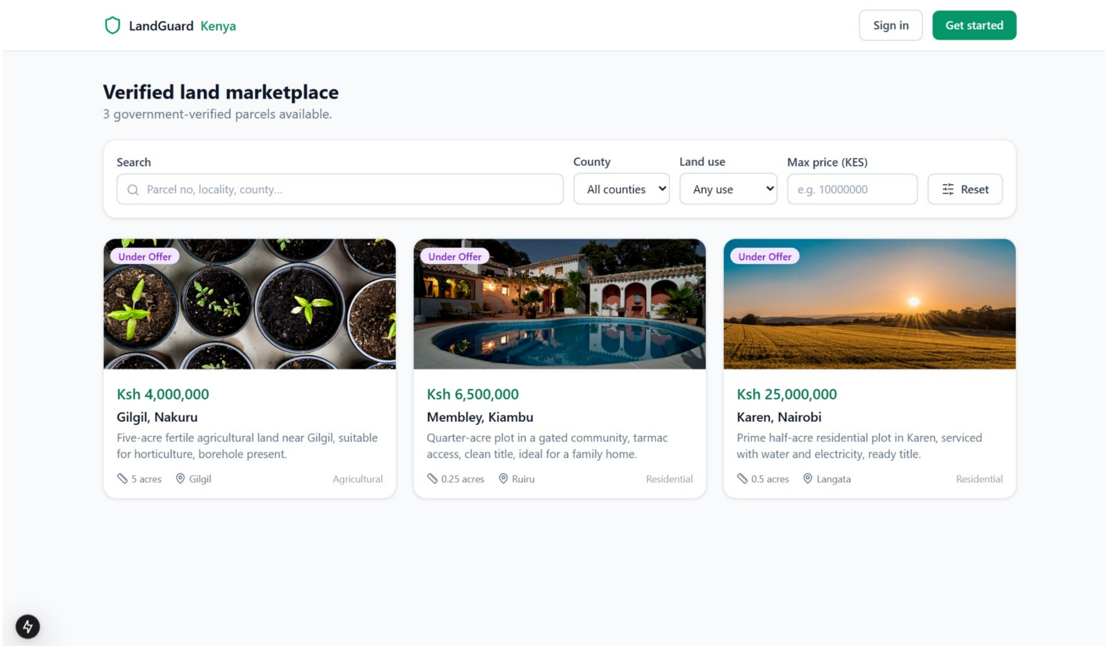

Figure 4.12: Verified land marketplace

The figure above shows the marketplace where buyers browse verified parcels and filter by county, land use and price. Every listing shown here has passed all three verification stages.

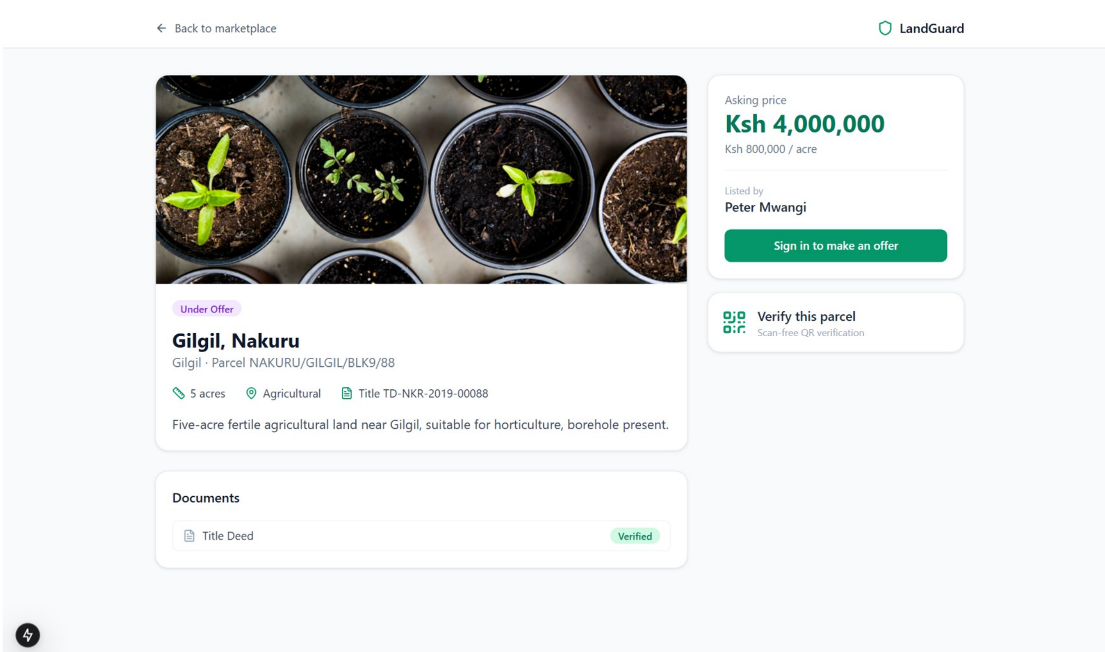

Figure 4.13: Parcel details with verification trail

The figure above shows a parcel page with its price, documents and the verification trail through administrator, government and survey approval, so a buyer can see exactly who verified the land and when.

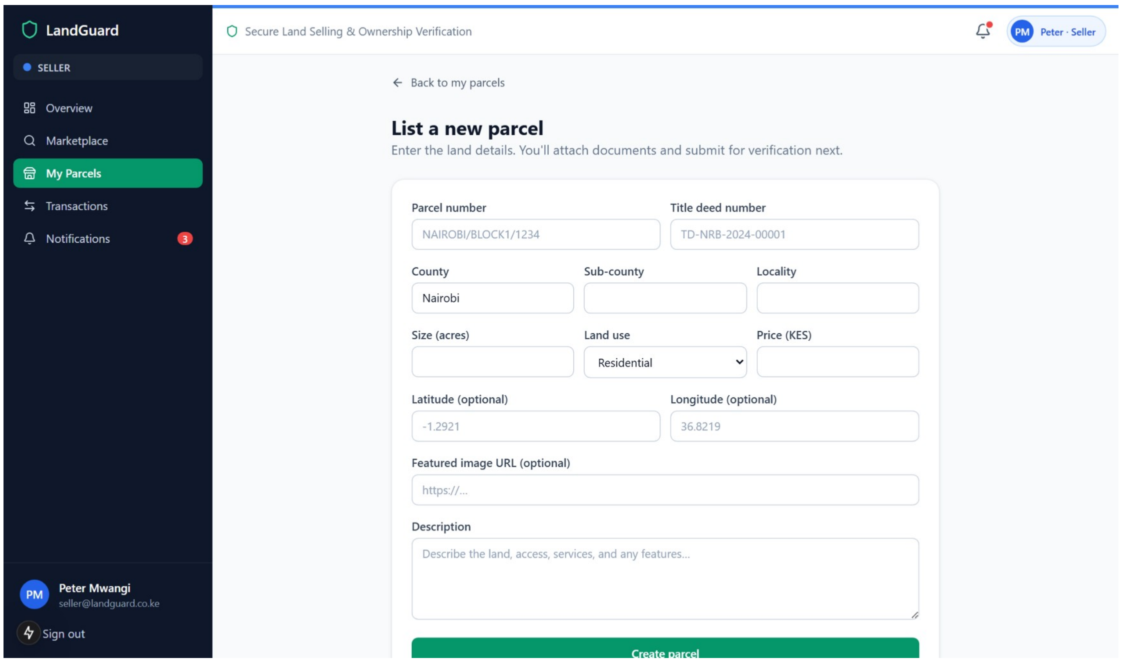

Figure 4.14: Seller listing a new parcel

The figure above shows the seller form for registering a parcel. Validation runs before the record is saved, and the parcel remains a draft until documents are attached and it is submitted for verification.

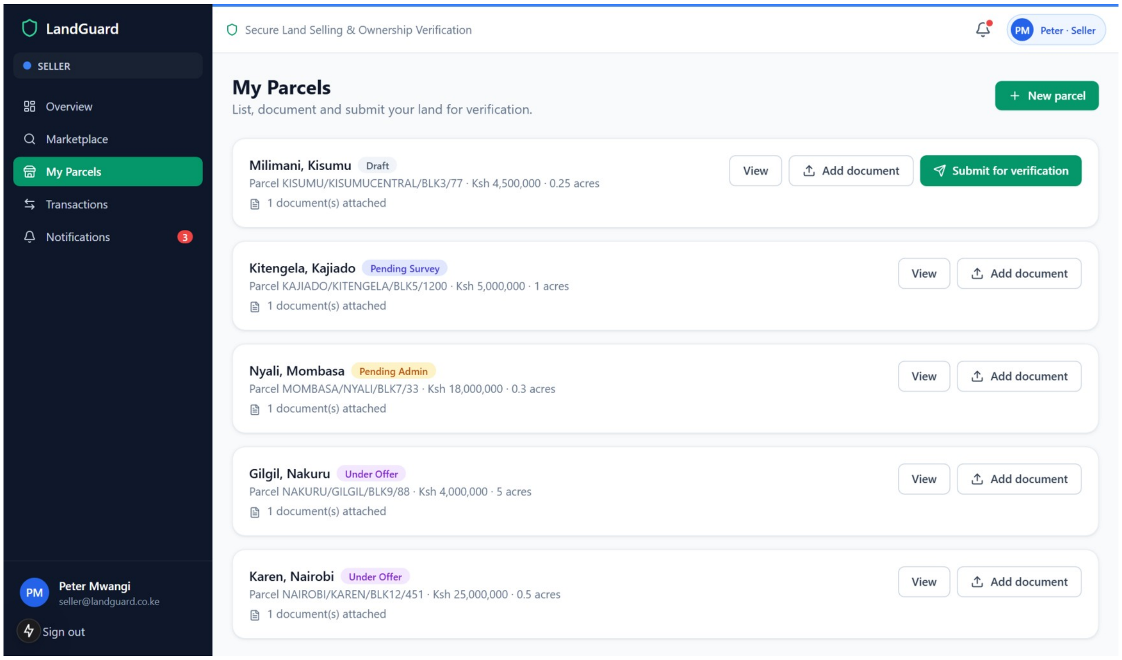

Figure 4.15: Seller parcel management and document upload

The figure above shows the seller dashboard where documents are attached and a parcel is submitted for verification. Submission triggers the fraud scan; a high-risk parcel is blocked on the spot with its risk score displayed.

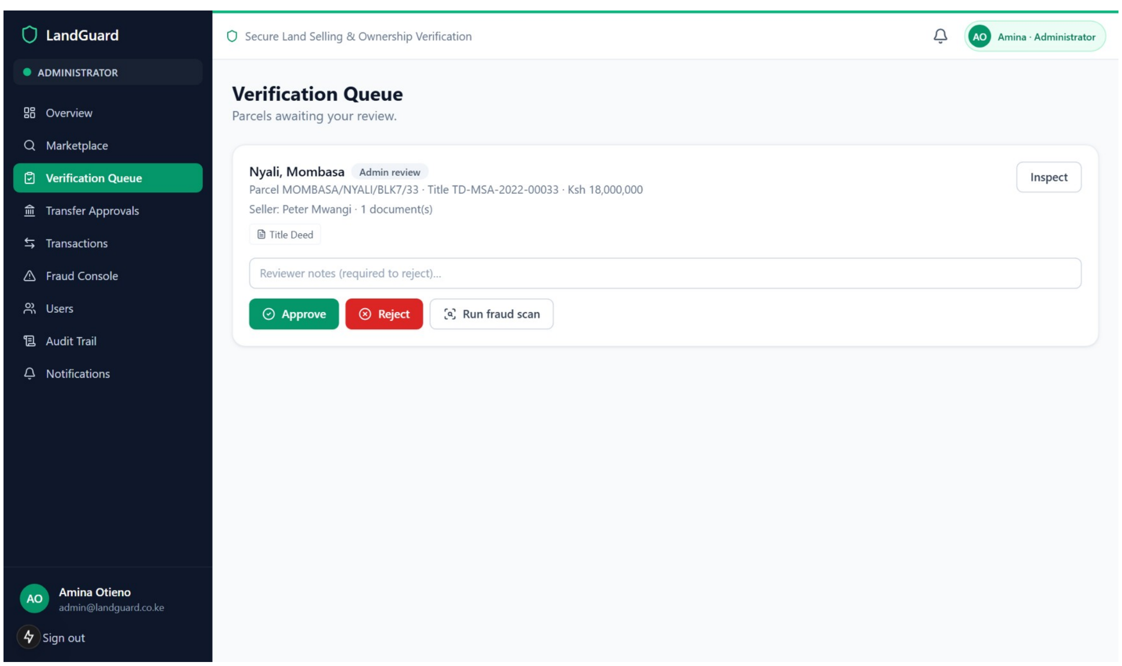

Figure 4.16: Verification queue

The figure above shows the review queue as seen by an administrator. Each entry shows the parcel, its seller, attached documents and any fraud warnings, with approve, reject and fraud-scan actions.

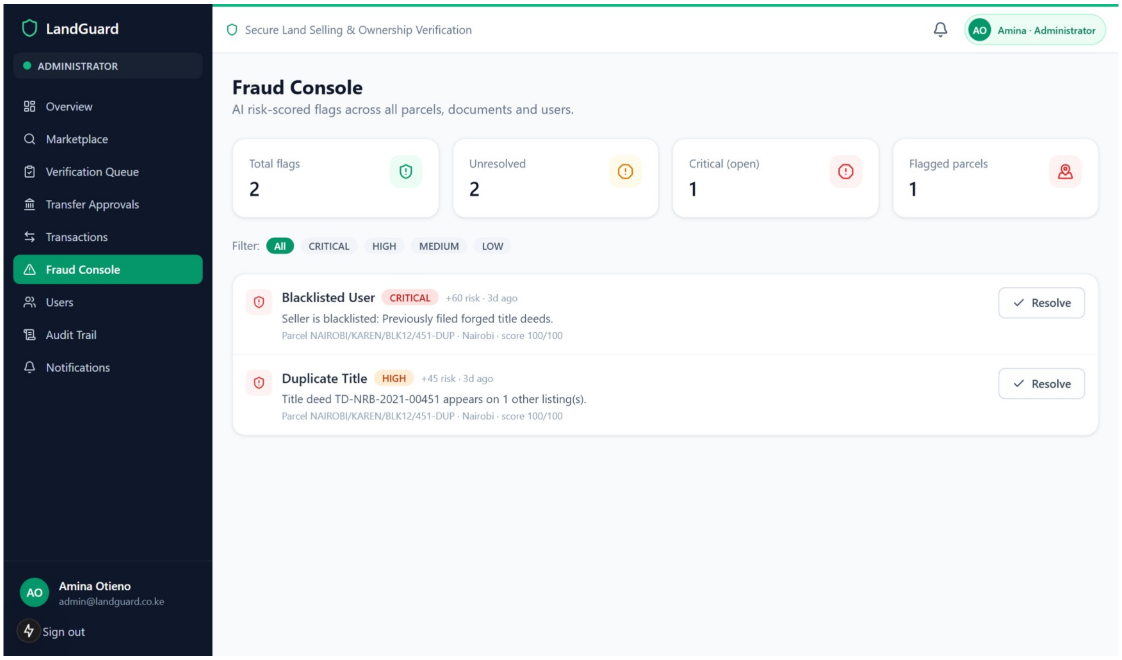

Figure 4.17: Fraud console

The figure above shows the fraud console listing risk-scored flags. Here a duplicated title deed and a blacklisted seller have pushed a parcel to a score of one hundred and it has been flagged automatically.

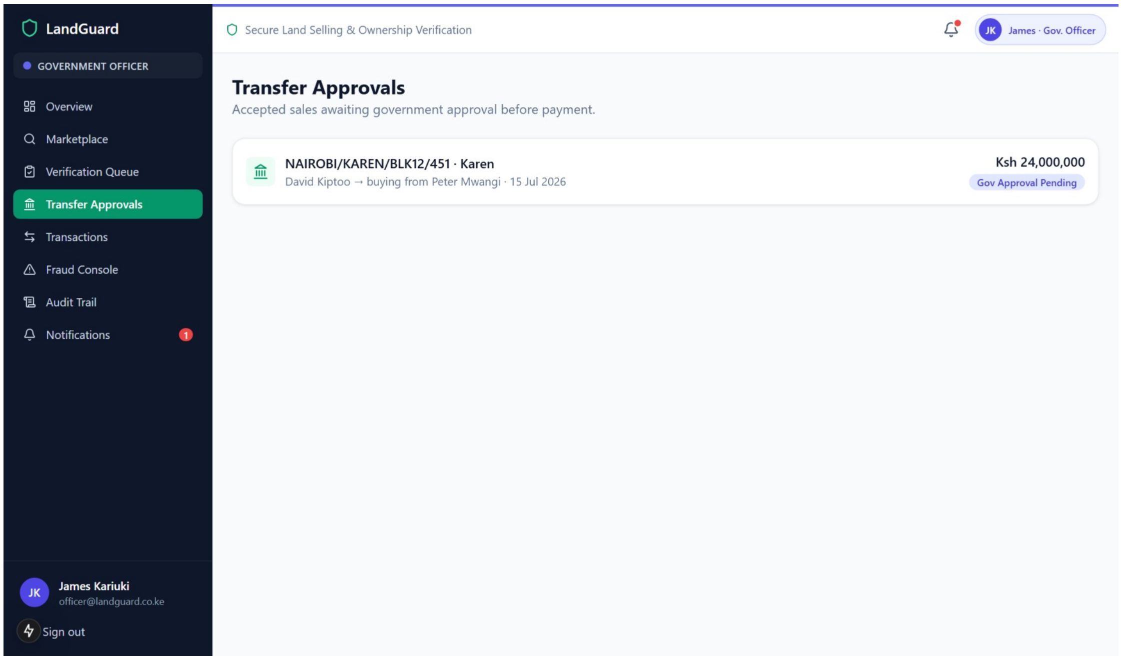

Figure 4.18: Government transfer approvals

The figure above shows the government officer's approval queue. Accepted offers wait here, and payment stays locked until the officer approves the transfer.

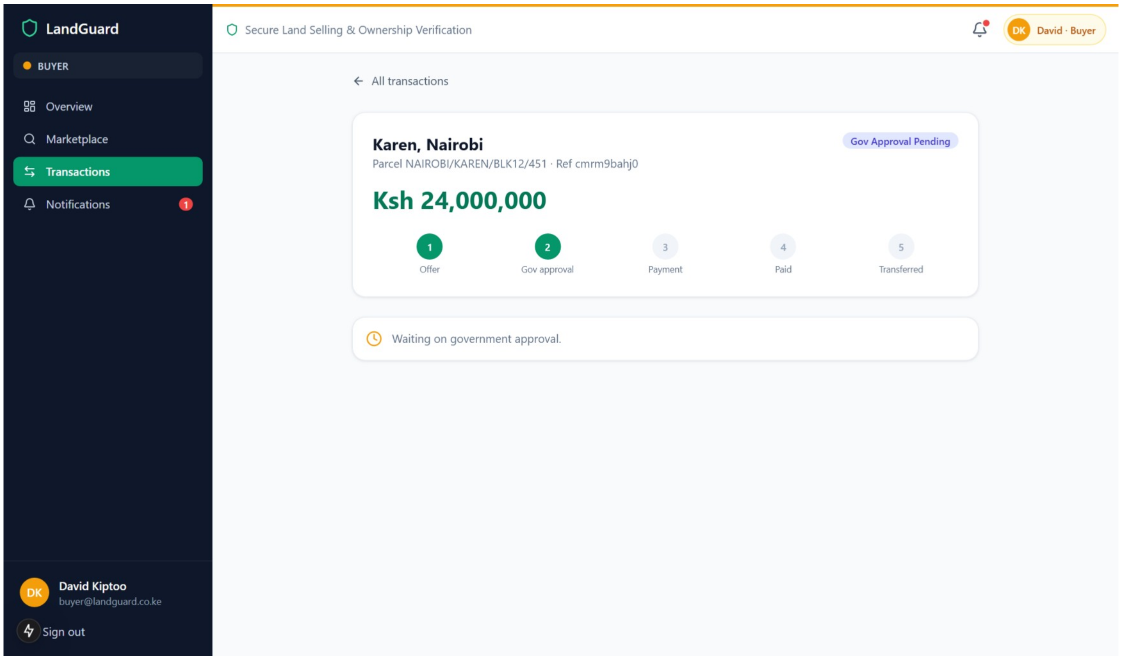

Figure 4.19: Transaction progress and payment

The figure above shows a transaction page with the five-step progress bar from offer to transfer. The buyer selects M-Pesa, card or bank transfer; on success the certificate panel appears on this page.

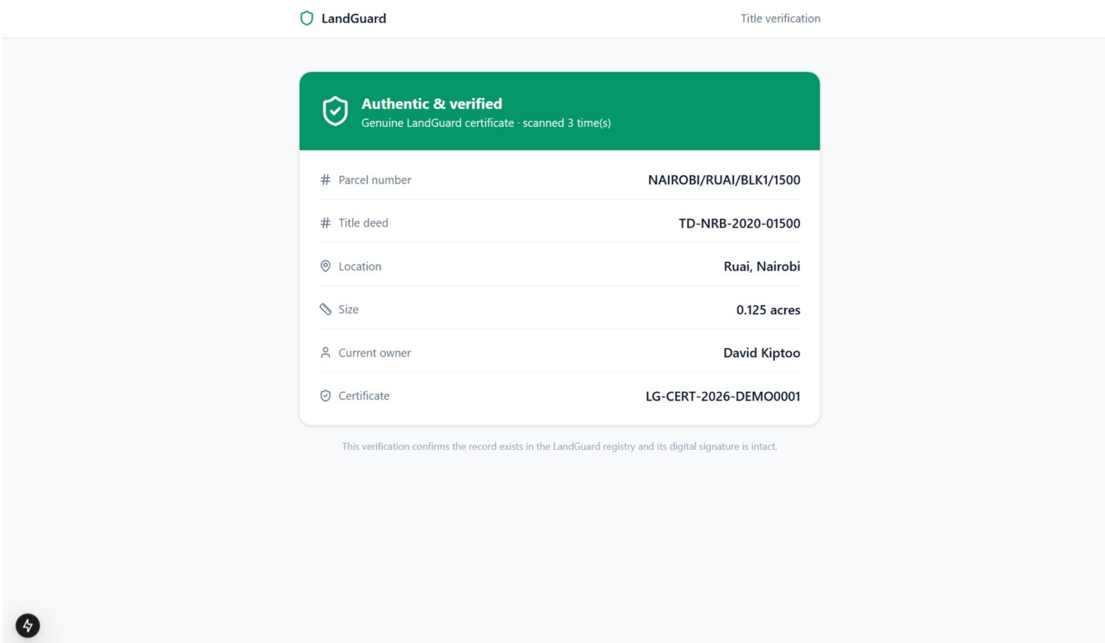

Figure 4.20: Public QR title verification

The figure above shows the public verification page opened from a certificate QR code. It confirms the record is genuine, shows the parcel, title number and current owner, and requires no account.

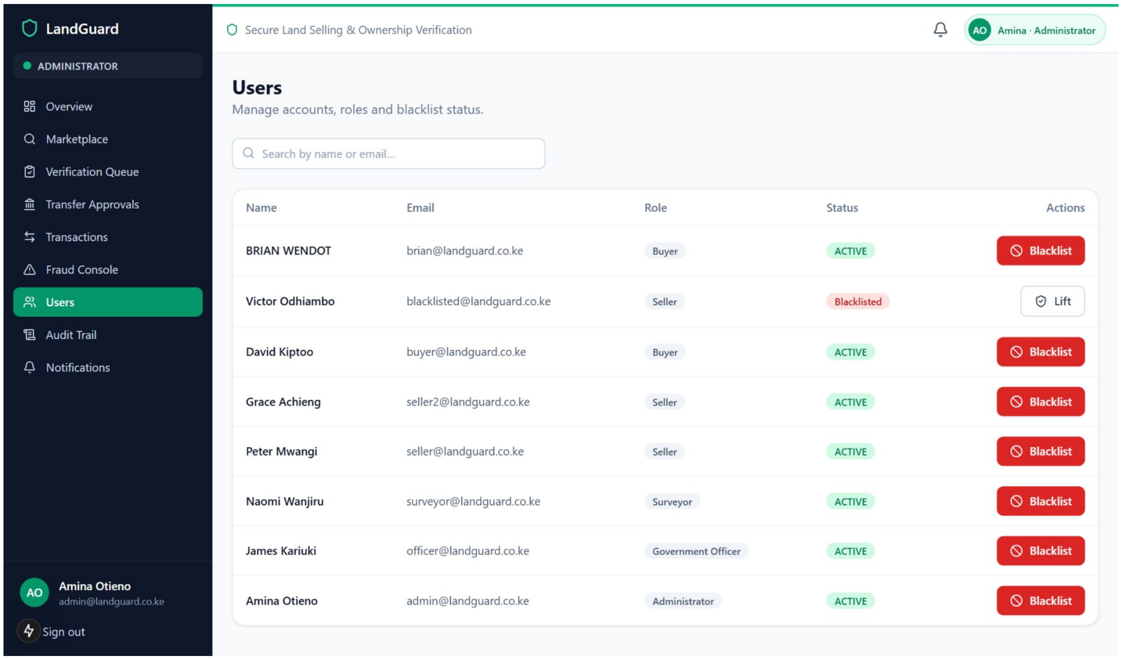

Figure 4.21: User management and blacklisting

The figure above shows the administrator's user management screen, including the blacklist action that feeds directly into the fraud engine.

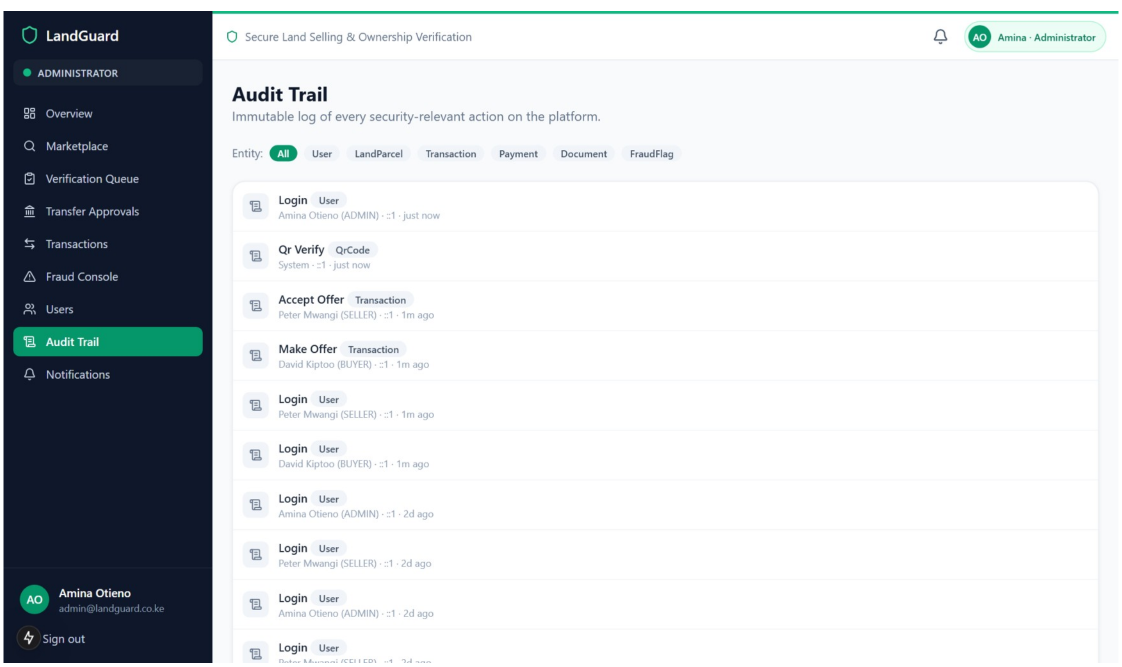

Figure 4.22: Audit trail

The figure above shows the audit trail, the append-only record of logins, submissions, decisions, payments and administrative actions used for accountability and dispute resolution.

4.9 Hardware and Software Specifications

Table 4.7: Hardware specification

Item	Minimum specification
Development computer	64-bit processor, 8 GB RAM, 10 GB free disk space.
Server (production)	2 vCPU, 4 GB RAM, 20 GB disk, Linux or Windows.
Client device	Any smartphone, tablet or computer with a modern browser.

Table 4.8: Software specification

Software	Version used
Node.js	20 or later (developed on 24)
Python	3.10 or later (developed on 3.14)
Next.js / React	15 / 19
Express	4
Prisma ORM	6
SQLite / PostgreSQL	Development / production databases
Flask	3

4.10 Description of Testing and Results

4.10.1 Testing Strategy

The system was tested at four levels. Unit testing exercised individual components, most importantly the fraud classifier, which was evaluated on a held-out test set. Integration testing exercised the interface endpoints against a seeded database. System testing executed complete end-to-end scenarios through the running platform, including a full purchase from offer to certificate. User acceptance testing walked through each role's tasks using the demonstration accounts. Security and performance checks ran alongside these levels.

Table 4.9: Fraud model evaluation results (held-out test set, 800 samples)

Metric	Result
Accuracy	98.9%
Precision	99.6%
Recall	97.1%
F1 score	0.98
Confusion matrix	True positives 271, true negatives 520, false positives 1, false negatives 8

Table 4.10: System test cases and results

ID	Test case	Expected result	Actual result	Verdict
TC1	Register and log in as buyer	Account created; token pair issued	As expected	Pass
TC2	Login with wrong password	Rejected with error; no token issued	As expected	Pass
TC3	Buyer calls an admin-only endpoint	Denied with a permission error	Denied (403)	Pass
TC4	Submit parcel without title deed	Submission blocked with clear message	As expected	Pass
TC5	Submit parcel with duplicate title deed and blacklisted seller	Fraud scan blocks and flags the parcel	Score 100/100; blocked and flagged	Pass
TC6	Full verification pipeline (admin, officer, surveyor approve)	Parcel becomes listed with QR code	As expected	Pass
TC7	Pay before government approval	Payment refused	Refused with workflow error	Pass
TC8	Complete sale: offer, accept, approve, pay by card	Ownership transfers; certificate issued	Parcel SOLD; certificate LG-CERT-2026-CMRHX48D issued	Pass
TC9	Scan certificate QR without an account	Public page confirms authenticity and owner	As expected	Pass
TC10	Stop the fraud model service and rescan	Scoring falls back to the rule engine	Fallback engaged; score still 100	Pass
TC11	Tampered QR payload	Verification reports signature mismatch	Reported as possible forgery	Pass
TC12	Repeated login attempts beyond the limit	Requests rate-limited	Limited with a clear message	Pass

Table 4.11: Performance measurements

Measurement	Result
Interface health endpoint	3 to 4 milliseconds
Dashboard statistics endpoint	about 20 milliseconds
Marketplace search endpoint	under 10 milliseconds
Production page loads (home, login, marketplace, fraud console)	6 to 10 milliseconds each
Fraud model scoring round trip	well under the 2.5 second timeout, typically tens of milliseconds

4.10.2 Description of Results

All twelve system test cases passed. The two most important results are the fraud blocking test and the complete sale. In the fraud test, a parcel that reused an existing title deed number and belonged to a blacklisted seller was scored one hundred out of one hundred and blocked from submission, with both signals reported in plain language on the fraud console. In the complete sale, a buyer's offer moved through seller acceptance and government approval, payment settled, ownership transferred to the buyer, the ownership history recorded the change, and certificate LG-CERT-2026-CMRHX48D was issued and verified publicly through its QR code. The fallback test confirmed that fraud screening continues even with the model service switched off.

4.11 System Validation and Deployment

The system runs from a single command in development, which starts the web application, the interface and the fraud service together with a seeded demonstration registry, and a container configuration is provided that adds PostgreSQL for production. Switching the database from SQLite to PostgreSQL requires changing one provider line and one connection string, after which the migrations rebuild the identical schema. A deployment guide, a user manual and interface documentation accompany the source code.

CHAPTER FIVE: CONCLUSION AND RECOMMENDATION

5.0 Introduction

This chapter concludes the study, makes recommendations arising from it, and outlines the work that should follow.

5.1 Conclusion

The study set out to reduce land fraud by building verification, official approval, fraud scoring and controlled payment into a single platform. The developed system meets that goal in its tested environment: no parcel can be listed without passing document checks, a fraud scan and three human approvals; no payment can be made before a government officer approves the transfer; and every completed sale ends in an ownership record and a signed certificate that anyone can verify free of charge. Testing confirmed the behaviour end to end, including the deliberate blocking of a fraudulent listing and the completion of a genuine sale. The project demonstrates that the gaps between discovery, verification, approval and payment, where land fraud currently lives, can be closed with well-understood technology.

5.2 Recommendation

It is recommended that the platform be piloted with a county land registry so that verification decisions are grounded in official records rather than uploaded copies, and that the payment integrations be moved from sandbox to live credentials under an escrow arrangement. Institutions handling land disputes would also benefit from read access to the audit trail and ownership history, which provide a neutral record of who did what and when.

5.3 Future Works

i. Direct integration with the national land information system so registry searches happen inside the verification pipeline.

ii. An automated test suite wired into continuous integration to protect the workflow logic as the system grows.

iii. Map rendering of parcel boundaries in the web application using the stored coordinates.

iv. A mobile application sharing the same interface, and Kiswahili localisation for wider reach.

v. Retraining the fraud classifier on real, labelled case data as the platform accumulates history.

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APPENDICES

Appendix A: Source Code Availability

The complete source code of the system, including the web application, the business interface, the fraud scoring service, the database schema and all documentation, is hosted on GitHub at https://github.com/Ambroise2/landguard (to be published; see submission checklist). The repository includes a single-command setup that installs all services, prepares the database with demonstration data and starts the platform.

Appendix B: Demonstration Accounts

The seeded registry provides one account per role; the password for all of them is Password123!.

Table A.1: Demonstration accounts

Role	Email
Administrator	admin@landguard.co.ke
Government Officer	officer@landguard.co.ke
Surveyor	surveyor@landguard.co.ke
Seller	seller@landguard.co.ke
Buyer	buyer@landguard.co.ke

Appendix C: Setup Summary

First run: npm run setup (installs all three services, creates configuration files and seeds the database). Every run: npm run dev, or the provided one-click launcher, which starts the web application on port 3000, the interface on port 4000 and the fraud service on port 5001 together. A production build with instant page loads is available through npm run fast. Full instructions are in the deployment guide inside the repository.